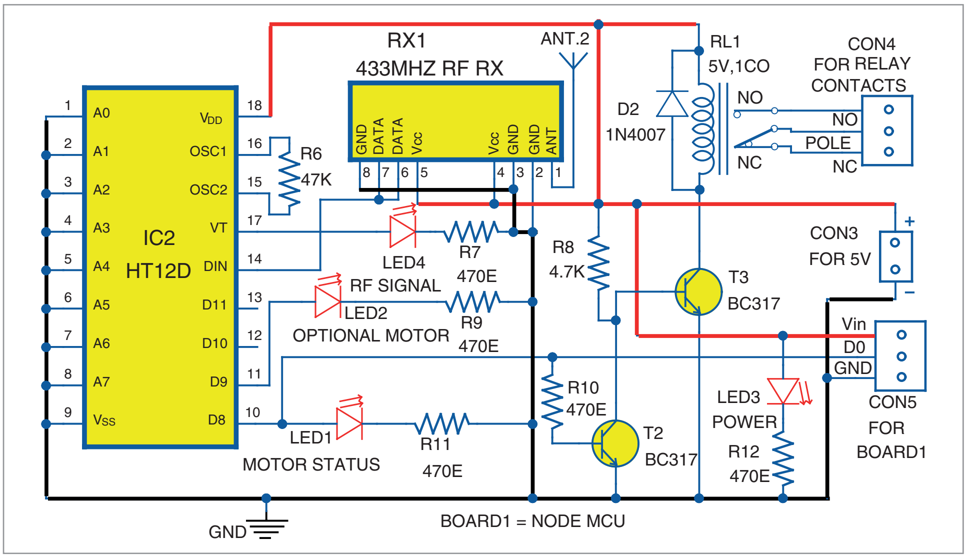

Fig. 3: Circuit diagram of receiver

Fig. 4: NodeMCU module

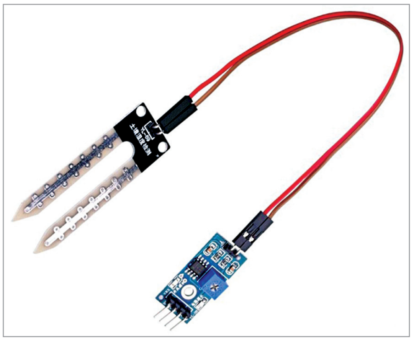

Fig. 5: Soil sensor module

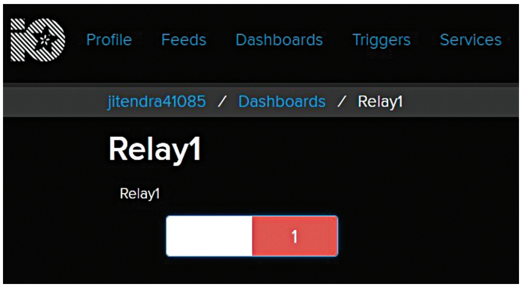

Fig. 6: Relay1 on

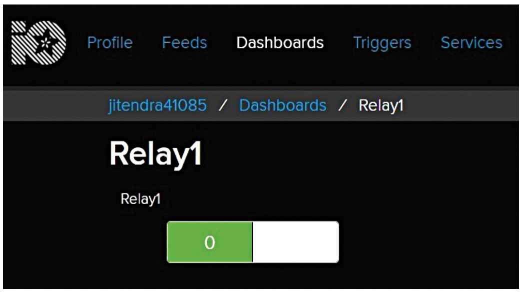

Fig. 7: Relay1 off

its collector low, which turns transistor T3 off. This de-energises relay RL1 and the pump connected to it is switched off.

Pump's switching on. When soil is dry, SS1 output pin D0 is high, which makes transistor T1 to conduct. So, collector voltage of T1 becomes low (0V), which is connected to pin AD8 of IC1. Transmitter TX1 transmits this signal, which is received by receiver RX1, decoded, and passed on to transistor T2. This makes transistor T3 to conduct and energise relay RL1, making the pump switch on.

Adafruit IoT platform

NodeMCU Wi-Fi is connected to Adafruit.io open source cloud service to provide real-time data online. Steps to organise the dashboard on Adafruit.io IoT platform are:

1. Open *https://io.adafruit.com* website. Create an account with a unique username. Note down this username as it will be used in Arduino code (program_soil.ino) later.

2. Generate a key and note down this key. This is a long, unique identifier that you use to authenticate any device using your account. You get one key per account, but you can revoke and regenerate your key any time.

3. After creating an account, you need to manage your feeds for publishing on the dashboard. (Feeds is basically a set of data that you can read or write, as specified in the program, as per your application.) In this project, we used Relay1 on the dashboard, as shown in Figs 6 and 7.

More details regarding configuration with Adafruit IoT and NodeMCU are given in 'Interfacing with Adafruit and NodeMCU' document file in the source code folder.

Software

Arduino IDE is used for programming the NodeMCU board (Board1). The Arduino code 'program_soil.ino' is to be uploaded into NodeMCU board before connecting it to the receiver circuit.

Connect the NodeMCU to PC/laptop and select proper COM port and board name from Tools menu of the Arduino IDE. The board name we used in this project is 'NodeMCU 1.0 ESP-12E Module.' The board name will depend on the model you are using.

Before compiling and uploading the source code, do not forget to include ESP8266WiFi.h, Adafruit, and MQTT libraries from Library Manager. Also, you need to add the 'package_esp8266com_index.json' package under Preferences in your Arduino IDE, as shown in Fig. 8.

Next, open program_soil.ino code from Arduino IDE and make some changes, such as Wifi SSID, password, Adafruit username, and key with your own credentials, as shown in the code below:

```
#include <ESP8266WiFi.h>
#include <Adafruit_Sensor.h>
```